

AI USAGE JOURNAL

Section 1: Tools Used

Google – This was used as the main research tool to analyse the key papers, give technical explanations, identify relevant literature and organise source material for the study notes.

ChatGPT – This tool was used to improve the above-mentioned explanations, improve the design of the quiz, debug the simulator and help restructure the technical content into clearer written material.

Google Forms – This tool was used to construct the adversarial quiz and to give automated feedback on the answers submitted in the quiz.

GitHub – This tool was used to organise the assignment in a clean manner and to consistently update the repository.

Jupyter Notebook – This tool was used to develop the annotated code walkthrough and to demonstrate the experiment made to explain saliency maps.

Streamlit and Python – These tools were used to develop the interactive simulator used to demonstrate saliency maps and input gradients.

Section 2: Important Prompts

Prompt 1: Explain what input gradients and saliency maps are in basic terms using simple analogies. Priority should focus on intuition rather than mathematics. Relate the explanation to the understanding of which parts of an image impact a neural network's decision.

Response: NotebookLM relates saliency maps to the important parts of the image that affect the model's prediction. To describe how gradients reveal sensitive parts of the input, analogies such as looking at a picture of a gazelle were used.

Impact on Project: This prompt helped convert gradient-based explanations into intuitive descriptions that can be easily understood.

Prompt 2: Explain the mathematical formulas that are used for input gradients, vanilla saliency maps, guided backpropagation, and SmoothGrad. Include equations and step-by-step algorithm descriptions.

Response: NotebookLM uses mathematical equations to explain each saliency method. This included explanations of how the gradient was calculated, how ReLU filtering was used in guided backpropagation, and the noise-averaging process in SmoothGrad.

Impact on Project: This directly formed the mathematical examples section of the study notes and supported the creation of the equations section for the quick references.

Prompt 3: Discuss the limitations and failure modes of saliency methods making use of the uploaded research papers. The limitations should include examples from Adebayo et al. (2018) and Kindermans et al. (2017).

Response: The response highlights that some saliency methods produce visualisations that are independent of both the model's weights and the data generation process, even when model parameters are randomized. It also highlights that gradient-based methods can fail under input transformations.

Impact on Project: This formed the failure mode section of the study notes.

Prompt 4: Fix this Streamlit saliency simulator error and improve comparison mode for CPU-only execution.

Response: The response identified the problems related to the handling of the gradients, the detachment of the input tensors and the context of the execution of the simulator on Streamlit. Multiple iterations were needed to stabilise the implementation

Impact on Project: This improved the final simulator design and led to a working CPU-compatible interactive tool with comparison mode.

Prompt 5: Explain the concept of saliency from biological vision using Itti et al., then connect it to deep learning saliency maps.

Response: The response describes how biological vision uses pre-attentive techniques to show important areas in a scene. This was linked to computational saliency maps in neural networks.

Impact on Project: This formed the foundations for the biological foundations section of the study notes.

Section 3: Reflection Questions

Hardest Concept to Explain

The hardest concept to explain in this project was how gradient-based attribution methods such as vanilla saliency and guided backpropagation actually relate to what the model “sees” an image. While the mathematics is based on partial derivatives, this does not translate naturally into intuition. The difficulty arose from bridging the gap between abstract and gradient values and human-interpretable visual importance. Even when analogies were used, the challenge of explaining why some pixels matter more than others still emerged.

AI that looked correct but was incomplete

The responses that were generated by NotebookLM initially appeared highly reliable when suggesting relevant papers and explanations. However, some responses included surveys that were not directly focused on input-space saliency methods. At first, these seemed suitable, but later I realised they lacked specificity for SmoothGrad and guided backpropagation. This was resolved by manually verifying each paper through the abstracts and PDFs, removing the sources that were deemed less relevant, and refining the literature to only focus on work that was related to saliency.

Where AI wasted time

A clear example of where the use of AI wasted time can be seen when debugging the Streamlit saliency simulator using ChatGPT. The initial changes that were proposed generated new errors that caused the simulator to not work correctly. Each iteration required restarting the simulator, which slowed down the progress significantly. The problem that needed to be solved took a long time since the suggestions that ChatGPT was outputting were sometimes partially correct but incomplete. This taught me that debugging code through AI still requires careful validation rather than direct implementation.

Where AI genuinely helped

AI has significantly accelerated the work done to be able to write the study notes. This was mainly due to the features that were offered by NotebookLM. Instead of manually reading and comparing the research papers, NotebookLM was used to summarise multiple research papers and extract key ideas related to saliency methods and their limitations. The most important contribution of this AI tool was to identify the relationships between SmoothGrad, vanilla saliency, and reliability issues. This allowed the possibility to structure the study notes and design the simulator more efficiently.

Remaining uncertainty

After completing the Learning Pack, I was still not confident in explaining the deeper biological connection between human visual attention techniques and modern gradient-based saliency methods. While I managed to explain Itti et al.'s model and its relation to computational saliency, the transition from biological attention processes to neural network gradients still feels conceptually abstract. This gap remains a challenge because it needs a good theoretical understanding of both neuroscience and deep learning interpretability.